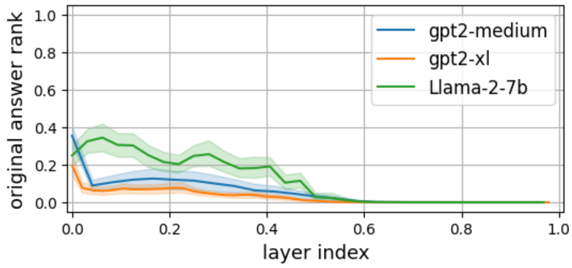
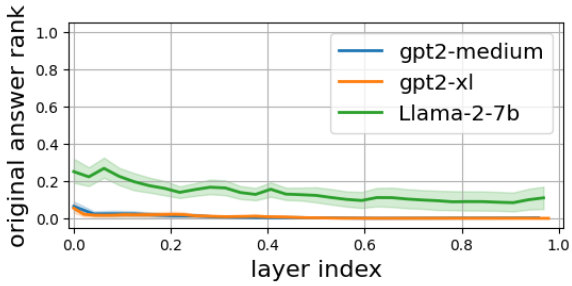


(a) Only correctly answered prompts



(b) Correctly and incorrectly answered prompts

Figure 20: FF_1's x_i ranks for the model's actual answers. The main difference between the two figures is that when the models answer incorrectly, they mostly answer with function words, such as "a" and "the", which seems to have a constant rank from the earlier layers, while answering the actual factual answers changes the answers' rank gradually in the first half of the layers. The meaning of these graphs is that during fine-tuning, the embeddings injected into the models' weights are those of x_i.

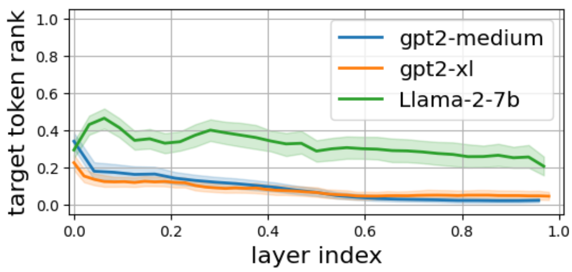


Figure 21: FF_1's x_i ranks for the edits' targets, reflecting the intermediate predictions the models give to them at each layer.

Throughout our study, this is the only result we found to be affected by the distinction whether the models answer the original prompt correctly or incorrectly.

Similarly, when we plot the ranks of the target token for each layer, we only observe the forward passes' intermediate probability for the target token (Figure 21).

G The Gradual Change of the VJPs Across Layers

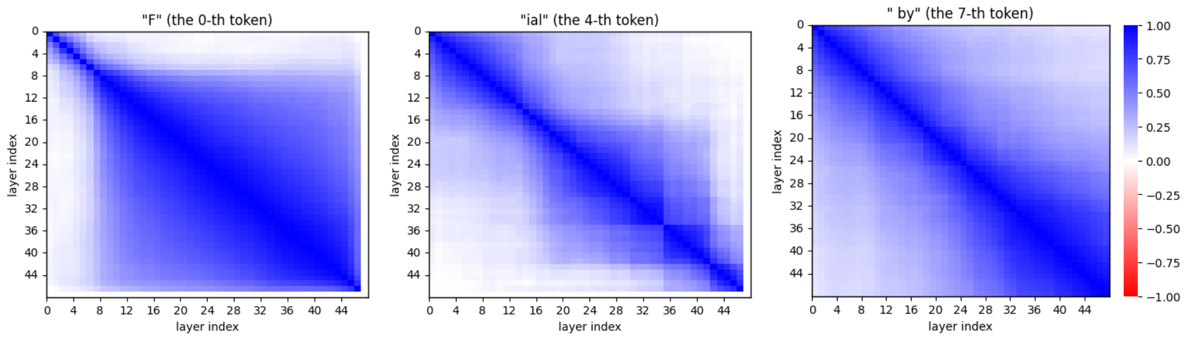
The results in Section F.2 reveal the presence of the embedded target within the MLP's FF_2 VJPs. To further extend this examination, we measure the similarity of the VJPs across different layers. Specifically, for two layers i and j, we calculate the cosine similarity between their VJPs for given token t, denoted as: $cosine(\delta_t^i, \delta_t^j)$. In particular, we are interested in the FF_2's VJPs since during the backward pass these VJPs are read by each FF_2 matrix from the residual stream. This allows us to explore how the gradient information propagates and evolves across layers.

We use the setup from Section F.2, with GPT2-xl, and visualize the results using a heat-map for each token. For the t token, the (i, j) cell of its map correspond to $cosine(\delta_t^i, \delta_t^j)$. An example for a single prompt is shown in Figure 22.

The VJP of the residual stream exhibits a cosine similarity score of approximately 0.4 over a sequence of 3-5 blocks for all tokens across various prompts.

Specifically, for the last token of each prompt, we calculate the averaged cosine similarity across 100 distinct edits. As shown in Figure 23, the cosine similarity score exceeds 0.7 across 10 blocks, indicating a strong alignment of VJPs between layers. Even when comparing the first layer (layer 0) to the last (layer 47), we observe an average cosine similarity above 0.08, which is relatively high score compared to random vectors in an embedding space of size 1600.

These results illustrate how the information from the initial VJP, originating in the model's final layer, permeates earlier layers and how gradually each layer modifies the residual VJP.



(a) first subject token (b) last subject token (c) the prompt last token

Figure 22: The cosine similarity between the residual VJPs (FF_2s' VJPs) of GPT-2 xl is examined. The provided maps are generated from editing the prompt "Ferrari Mondial, created by" with the target "Nintendo".

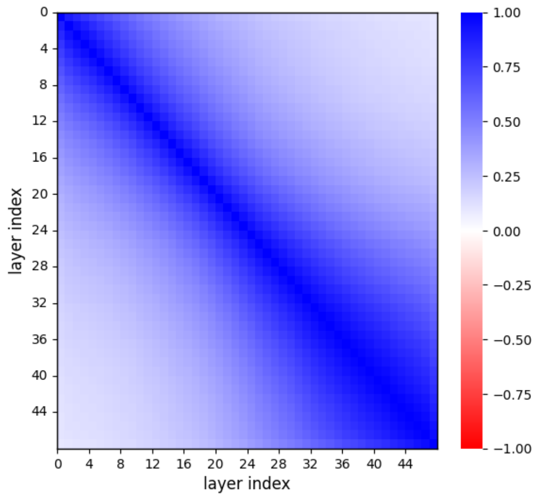


Figure 23: The cosine similarity between the residual VJPs of all GPT2-xl layers for the last token of a given edit, averaged across 100 distinct edits, shows a consistent alignment across layers.

H Normalized Logit Lens

We acknowledge the sensitivity of the Logit Lens to low-norm vectors. With vectors with close to zero norms, the tokens LL project tokens that resemble the projection of the zero vector. In Section 6, we discussed that certain VJPs, δ_i, exhibit low norms. We hypothesize that the norms of the VJPs reflect which parts of speech and layers the backward pass tries to edit more. We were interested in observing which tokens could be projected from gradients if we isolate the influence of these low norms. Our investigation reveals that normalizing the projected vectors before applying the Logit Lens can be an appropriate solution to the variance in VJPs' norms. We named this method "Normalized Logit Lens".

A place we consider using this normalization is in creating the tables of Appendix E. In Figure 24 and Figure 25 we share two examples that replicate the setup from Appendix E except we are using Normalized Logit Lens.

We provide this section to highlight the pattern we already mentioned in Section 6: that most of FF_2 gradients ranks the target token as one the most improbable token. We also want to suggest further work to consider using Normalized Logit Lens to examine low-norm vectors.

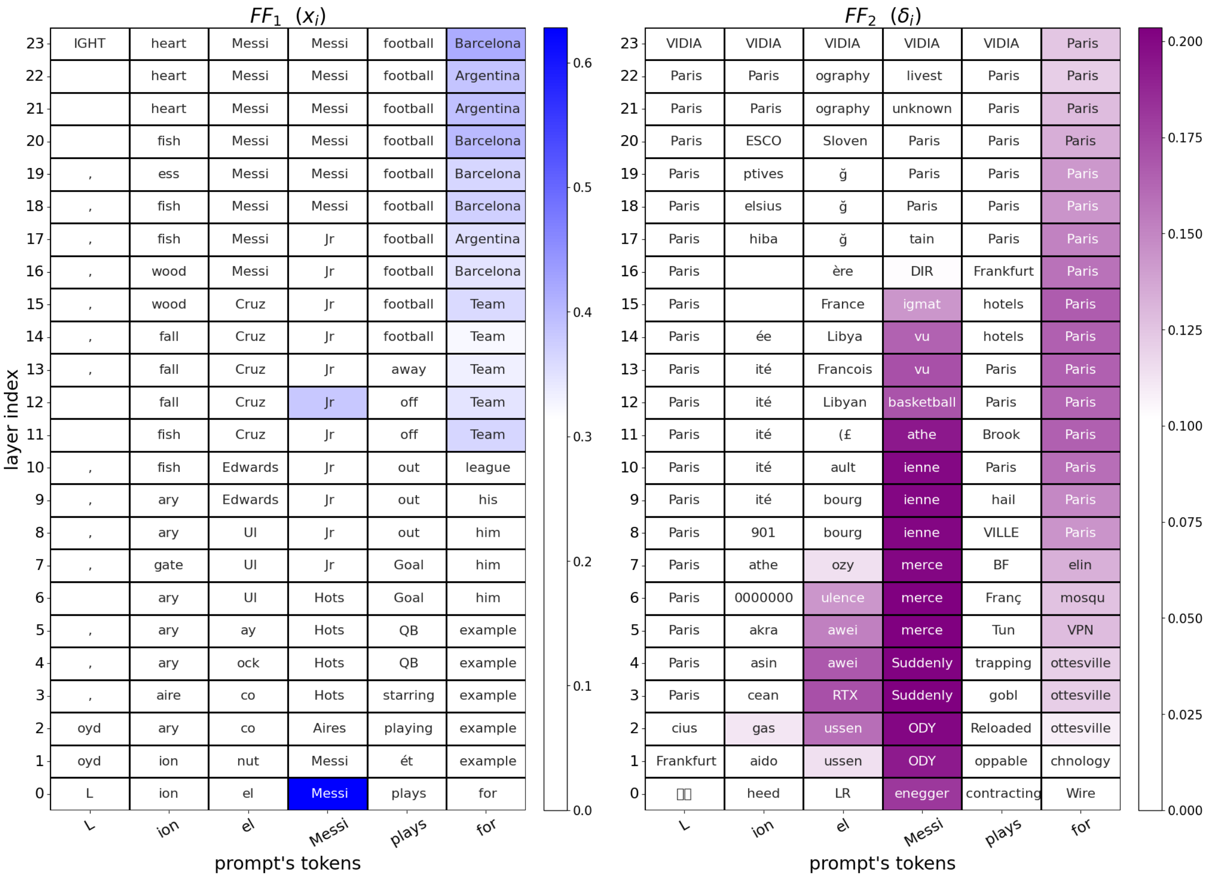


Figure 24: GPT2-medium MLP gradients projections with **Normalized Logit Lens**. Compared to the naive Logit Lens Figure 11, we see more cells in FF_2 that project concepts similar to the target token "Paris" and less unclear tokens, such as "VIDIA" (notice the last layers or the column of the token"ion"). The coloring is according to the original norm of each representative vector (before normalization).